FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Main)-2021

TEST DATE: 28-02-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A particle moves in x-y plane with velocity $\vec{v} = \alpha\hat{i} + \beta t\hat{j}$. The magnitude of tangential, normal and total acceleration at $t = \frac{\alpha\sqrt{3}}{\beta}$ are

(A) $\sqrt{2}\beta, \sqrt{2}\beta$ & β respectively

(B) $\frac{\sqrt{3}\beta}{2}, \frac{\beta}{2}$ & β respectively

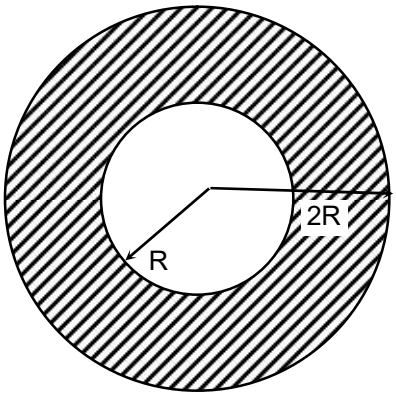
(C) $\frac{\beta}{2}, \frac{\beta}{2}$ & β respectively

(D) $\frac{\beta}{\sqrt{2}}, \frac{\beta}{\sqrt{2}}$ & β respectively

2. The potential energy of a certain spring when stretched through a distance x is 10 J. The amount of work (in Joules) that must be done on spring to stretch it through an additional distance x will be

(A) 10
(B) 20
(C) 30
(D) 40

3. The gravitational potential at the centre of a sphere of radius 2R having a concentric spherical cavity of radius R and net mass M is



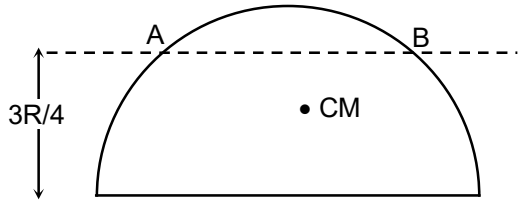
(A) $\frac{-3GM}{7R}$

(B) $\frac{-9GM}{14R}$

(C) $\frac{-5GM}{7R}$

(D) $\frac{-7GM}{14R}$

4. Assuming the centre of mass of a hemisphere to lie at a height $\frac{3R}{8}$ from the base, the radius of gyration of a solid hemisphere of mass M and radius R about an axis parallel to the diameter at a distance $\frac{3R}{4}$ from this plane is given by



(A) $\frac{3R}{\sqrt{10}}$

(B) $\frac{5R}{4}$

(C) $\frac{5R}{8}$

(D) $\sqrt{\frac{2}{5}}R$

5. A thin uniform rod of mass m moves translationally with acceleration a due to two antiparallel forces of lever arm ℓ. One force is of magnitude F and acts at one extreme end. The length of the rod is

(A) $\ell\left(1+\frac{F}{ma}\right)$

(B) $2\ell\left(1+\frac{F}{ma}\right)$

(C) $\frac{\ell}{2}\left(1+\frac{F}{ma}\right)$

(D) $\frac{ma\ell}{ma+F}$

6. When a metallic surface is illuminated with monochromatic light of wavelength λ, the stopping potential is $5V_0$. When the same surface is illuminated with light of wave length 3λ, the stopping potential is V_0. Then the work function of metallic surface is

(A) $\frac{hc}{4\lambda}$

(B) $\frac{hc}{5\lambda}$

(C) $\frac{hc}{6\lambda}$

(D) $\frac{hc}{12\lambda}$

7. Nuclei of radioactive element A are produced at rate 't²' at any time t. The element has decay constant λ. Let N be the number of nuclei of element A at any time t. At time $t = t_0$, $\frac{dN}{dt}$ is minimum. Then the number of nuclei of element A at time $t = t_0$ is

(A) $\frac{\lambda t_0^2 - 2t_0}{\lambda^2}$

(B) $\frac{\lambda t_0^2 - t_0}{\lambda^2}$

(C) $\frac{\lambda t_0^2 - 2t_0}{\lambda}$

(D) $\frac{\lambda t_0^2 - t_0}{\lambda}$

8. Two identical sound producing trains are travelling in same direction with velocity 20 m/s and 30 m/s respectively. They are approaching a hill. Find the ratio of reflected frequency heard by both train due to other train. Speed of sound = 300 m/s.

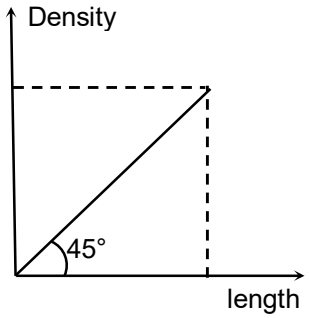
(A) $\frac{896}{891}$

(B) $\frac{891}{889}$

(C) $\frac{27}{28}$

(D) $\frac{32}{33}$

9. A string of length L of variable density is clamped between two rigid supports whose density variation is shown in figure. Find the time after which the disturbance created at one end will reach other end. (A = cross-sectional area of string, T = tension in string)



(A) $\frac{1}{3}\sqrt{\frac{A}{T}}L^{1/3}$

(B) $\frac{1}{3}\sqrt{\frac{A}{T}}L^{3/2}$

(C) $\frac{2}{3}\sqrt{\frac{A}{T}}L^{2/3}$

(D) $\frac{2}{3}\sqrt{\frac{A}{T}}L^{3/2}$

10. A carnot refrigeration cycle absorbs heat at 270 K and rejects at 300 K. If the cycle is absorbing 1260 kJ/min at 270 K, then work required per second is

(A) 1 kJ/sec
(B) 4 kJ/sec
(C) 2.33 kJ/sec
(D) 4.66 kJ/sec

11. Under an adiabatic process, the volume of an ideal gas gets quadrupled (4 times). Consequently the mean collision time between the gas molecule changes from τ_1 and τ_2. If $\frac{C_P}{C_V} = \gamma$ for this gas then a good estimate for $\frac{\tau_2}{\tau_1}$ is given by :

(A) $\left(\frac{1}{2}\right)^{\gamma}$

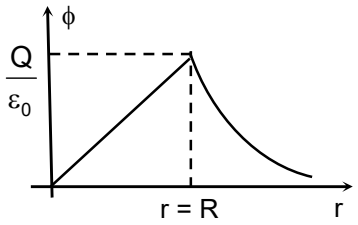
(B) $\left(\frac{1}{2}\right)^{\frac{\gamma+1}{2}}$

(C) $(2)^{\frac{\gamma+1}{2}}$

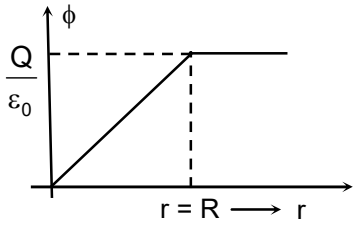
(D) $(2)^{\gamma+1}$

12. Electric flux (ϕ) as a function of distance (r) from centre of a uniformly charged solid sphere of charge (Q) and radius R, is best represented by

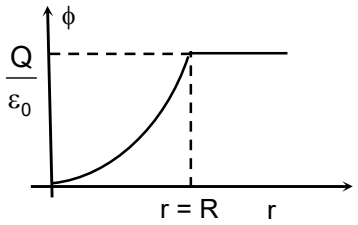
(A)



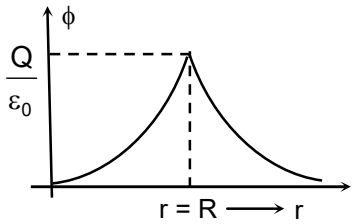
(B)



(C)

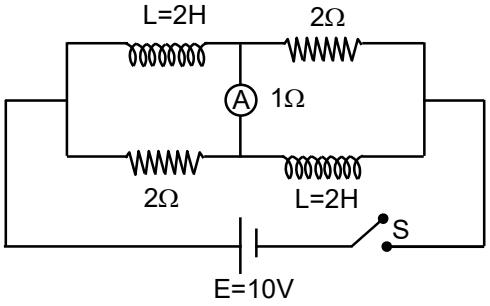


(D)



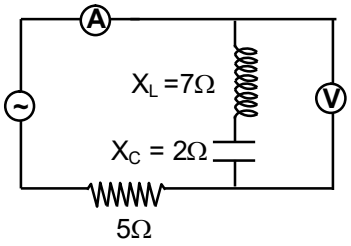
13. Initially all the inductors have zero current. At t = 0, S is closed. If at t = 0 current through ammeter is I_1 and after long time current through ammeter is I_2. Find $\frac{I_1}{I_2}$.

(A) 0.1
(B) 0.2
(C) 1
(D) 2



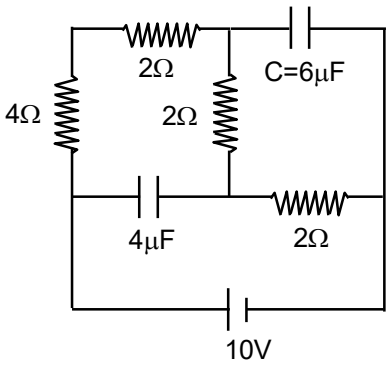
14. In the figure shown supply voltage is V = 10 sin ωt in volts. The reading of the hot wire voltmeter is (Assume both hot wire ammeter and hot wire voltmeter is ideal)

(A) 5 V
(B) $5\sqrt{2}$V
(C) 10 V
(D) $10\sqrt{2}$V



15. In the circuit shown in figure, battery is ideal. Find the potential difference across C = 6μF.

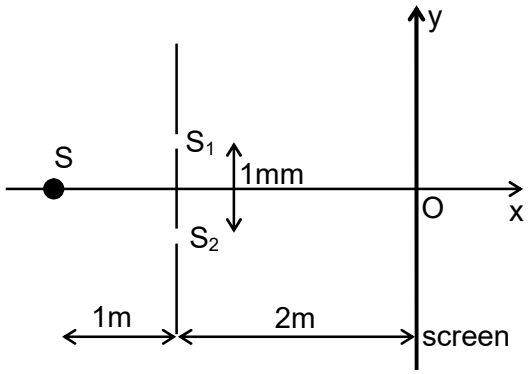
(A) 2 V
(B) 4 V
(C) 6 V
(D) 8 V



16. A spherical soap bubble is blown such that its radius increases at the constant rate of $\frac{dr}{dt} = \alpha$. Surface tension of the film is σ. When radius of the bubble is r, power required to increase the radius of the bubble is

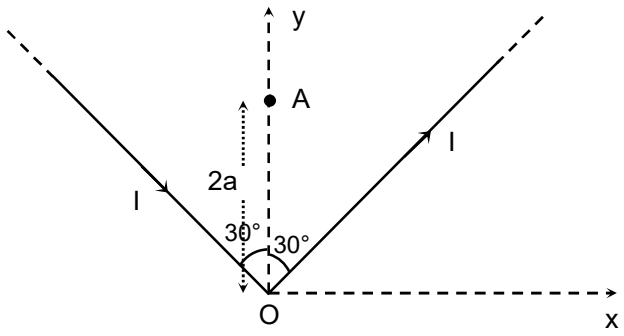
(A) $2\pi r\sigma\alpha$
(B) $4\pi r\sigma\alpha$
(C) $8\pi r\sigma\alpha$
(D) $16\pi r\sigma\alpha$

17. In a young's double slit experiment set up source S of wavelength λ = 500 nm illuminates two symmetrically located slits S_1 and S_2. The source S oscillates about its shown position parallel to the screen according to the equation y = (0.5 mm) sin (πt). Find least value of time (t) at which its intensity becomes maximum at a point on the screen that is exactly in front of slits S_1. (where t is time in second. Distance are as marked in the figure).



(A) $\frac{1}{3}$
(B) $\frac{1}{6}$
(C) $\frac{1}{12}$
(D) $\frac{1}{24}$

18. An infinite current carrying wire having current I is bent in V-shape, lying in x-y plane as shown in figure. Intensity of magnetic field at point A will be (OA = 2a)



(A) $\frac{\mu_0 I}{4\pi a}\left(2+\sqrt{3}\right)\hat{k}$
(B) $\frac{\mu_0 I}{4\pi a}\left(2-\sqrt{3}\right)\hat{k}$

(C) $\frac{\mu_0 I}{4\pi a}\left(2+\sqrt{3}\right)\left(-\hat{k}\right)$

(D) $\frac{\mu_0 I}{4\pi a}\left(2-\sqrt{3}\right)\left(-\hat{k}\right)$

19. Relation between velocity, displacement x (from mean position) and acceleration of a particle in a dissipative medium is given by $\frac{3d^2x}{dt^2}+\frac{5dx}{dt}+4x=0$. Then time in which its amplitude will be reduced to half of its initial value is

(A) $\frac{3}{5}\ln 2\,\text{sec}$

(B) $\frac{3}{5}\ln 5\,\text{sec}$

(C) $\frac{6}{5}\ln 2\,\text{sec}$

(D) $\frac{6}{5}\ln 5\,\text{sec}$

20. In an x-ray tube, if the accelerating potential is changed, then

(A) the frequency of characteristic x-rays of a material will get changed

(B) number of electrons emitted will change

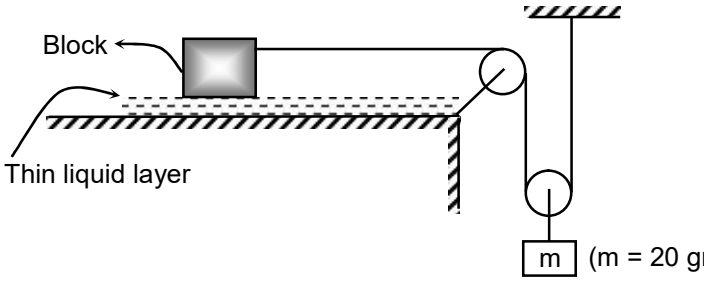
(C) the difference between λ_0 (minimum wavelength) and $\lambda_{k\alpha}$ will get changed

(D) difference between $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ will get changed.

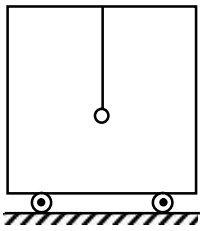
SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A cubical block of side $10\sqrt{10}$ cm is connected as shown in the figure. The block is placed on a thin layer of liquid of thickness 0.2 mm. If load moves downward with constant speed of 2 cm/sec after the system is released. The coefficient of viscosity of liquid is $\frac{x}{1000}$ P_a-sec, then x is (Take g = 10 m/s^2)

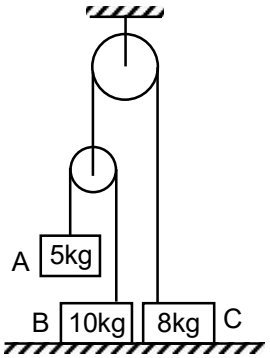


22. Bob of pendulum of length 'ℓ' has mass 'm' and cart having mass '2m'. If time period of small vertical oscillations in the plane of paper is $2\pi\sqrt{\frac{K\ell}{g}}$ then 3K is (all surfaces are smooth)



23. A hydrogen atom is in its excited state with energy – 1.51 eV. The angular momentum of the electron in this state is $\frac{xh}{2\pi}$, then write the value of x.

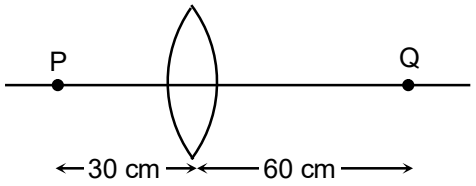
24. In the following arrangement the system is initially at rest. The 5 Kg block is now released. Assuming the pulley and string to be massless and smooth. Find the acceleration of block C of mass 8 kg in m/s^2. (Take g = 9.8 m/s^2)



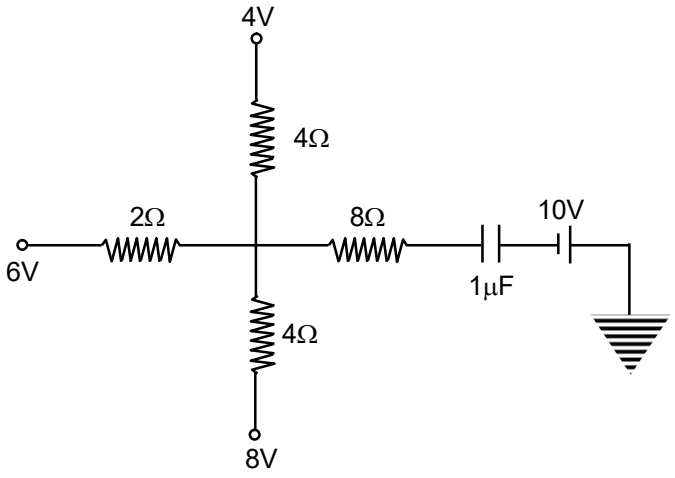
25. The ionization energy of hydrogen atom is 13.6 eV. Hydrogen atoms in the ground state are excited by electromagnetic radiation of energy 12.09 eV. How many spectral lines will be emitted by the hydrogen atoms?

26. Assuming that 200 MeV of energy is released per fission of uranium atom, then the number of fission per second required to release 1 Kilowatt power is N × 10^{12}. Find N

27. Two point objects P and Q are kept on the principal axis of a convex lens as shown. Image of both the objects is formed at same position. Find the focal length of lens in cm.



28. Figure shows a network of a capacitor and resistors. The charge on capacitor in steady state is (in μc)



29. The forward biased resistance of a PN junction diode is 10 Ω and its knee point is 0.7 Volt. The diode is connected across a load resistance of 500 Ω and AC supply of rms voltage $10\sqrt{2}$ Volt. Find the peak voltage across the load.

30. In a telescope focal lengths of objective and eye piece are 48 cm and 4 cm respectively. It is used to focus a distant object with final image at infinity. If object subtend an angle 2° on eye, find angular width of final image in degrees.

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. For x-axis, wave function (ψ) can be written as

$$\psi = A\sin\left(\frac{2\pi x}{\lambda}\right)$$

Thus, Schrodinger wave equation along x-axis can be written as:

(A) $\frac{d^2\psi}{dx^2} - \frac{4\pi^2\psi}{\lambda^2} = 0$

(B) $\frac{d^2\psi}{dx^2} + \frac{2\pi\psi}{\lambda} = 0$

(C) $\frac{d^2\psi}{dx^2} + \frac{4\pi^2\psi}{\lambda^2} = 0$

(D) None of these

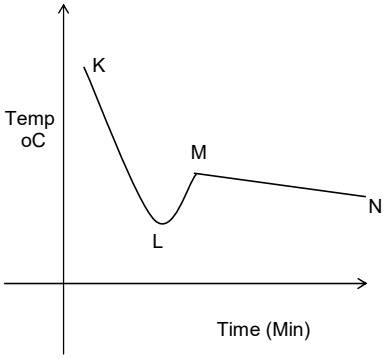
32. The curve shown results when a liquid is cooled. What temperature is closest to the freezing point of the liquid?

(A) L

(B) M

(C) $\frac{L+M}{2}$

(D) $\frac{M+N}{2}$



33. Which of the process happen during the discharging of a lead storage battery

1. H_2 gas is produced.
2. PbO_2 is converted to $PbSO_4$.
3. The density of the electrolyte solution decreases.

(A) 1 only

(B) 2 only

(C) 1 and 3 only

(D) 2 and 3 only

34. During the complete combustion of methane (CH_4), what change in hybridization does the 'C' atom undergo?

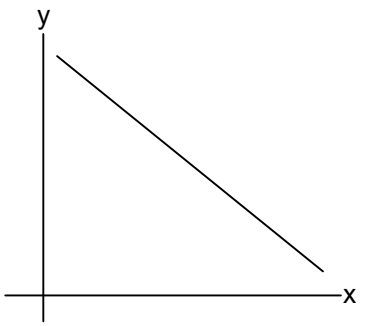
(A) sp^3 to sp

(B) sp^3 to sp^2

(C) sp^2 to sp

(D) sp^2 to sp^3

35. A student analysed the data from a zero order reaction and obtained the graph as shown below. What labels should be attached to the x and y axes, respectively



(A) time, concentration
(B) time, 1/concentration
(C) time, $\ell n\,(\text{concentration})$
(D) 1/time, concentration

36. The gas phase reaction shown below is endothermic as written. Which change (s) will increase the quantity of $CH_3 - CH = CH_2$ at equilibrium

$$\left[CH_3 - CH = CH_2 \rightleftharpoons \begin{array}{c} C \\ / \quad \backslash \\ C \text{—} C \end{array} \right]$$

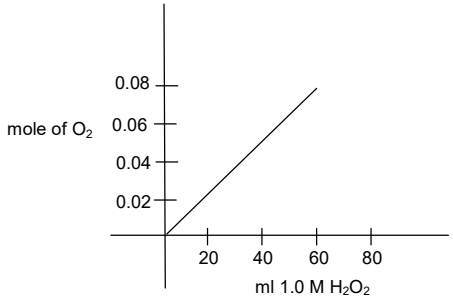
(I) Increase the temperature
(II) Increase the pressure
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

37. A typical polythene grocery bag weighs 12.4 gm. How many metric tons of CO_2 would be released in to the atmosphere if the 102 billion bags used in one year in India were burnt? (Given 1 kg of plastic will produce 6 kg of CO_2.)
(Given 1 metric ton = 1000 kg)
(A) 4.52×10^4
(B) 1.99×10^6
(C) 7.59×10^6
(D) 3.98×10^9

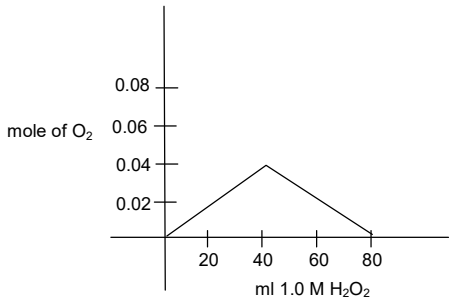
38. A sample of 0.040 mole hypochloride ion is treated with varying amount of 1.0 M aqueous H_2O_2. Which graph represents the amount of O_2 gas that is evolved according to the chemical reaction shown?

$$OCl^-(aq) + H_2O_2(aq) \longrightarrow Cl^-(aq) + H_2O(\ell) + O_2(g)$$

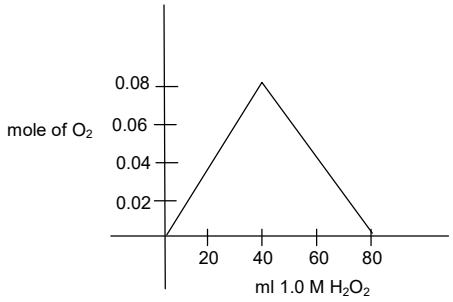
(A)



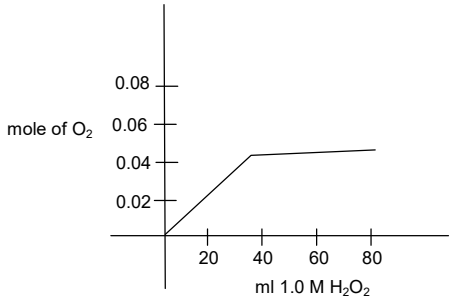
(B)



(C)

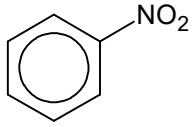


(D)

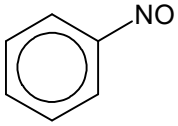


39. What is the product of the reaction of benzene with a solution of nitric acid in sulphuric acid at 50°C?

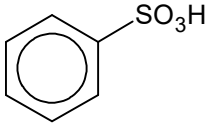
(A)



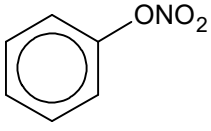
(B)



(C)

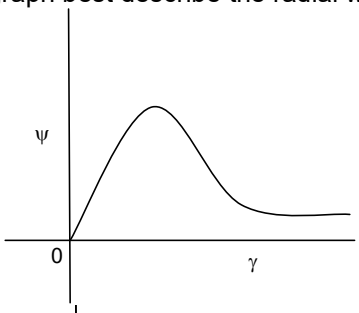


(D)

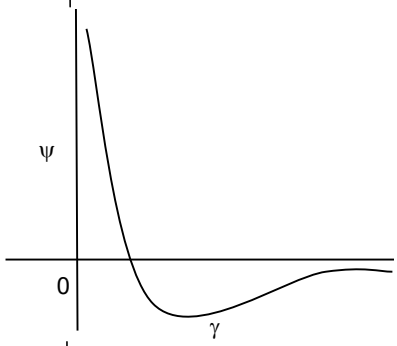


40. Which graph best describe the radial wave function of a 2p-orbital?

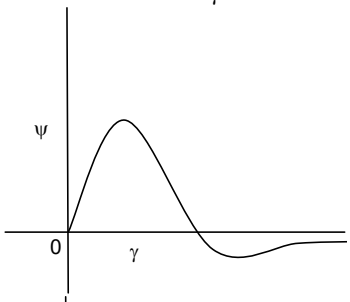
(A)



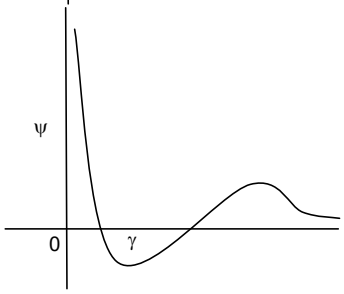
(B)



(C)



(D)



41. In hydrogen atom, which transition requires the greatest input of energy?

(A) $n = 7 \rightarrow n = 3$

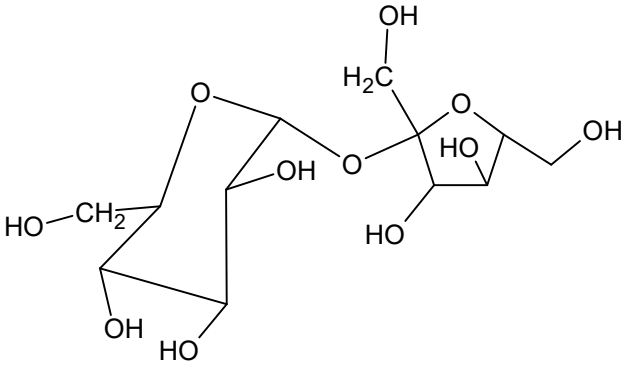
(B) $n = 2 \rightarrow n = 1$

(C) $n = 3 \rightarrow n = 7$

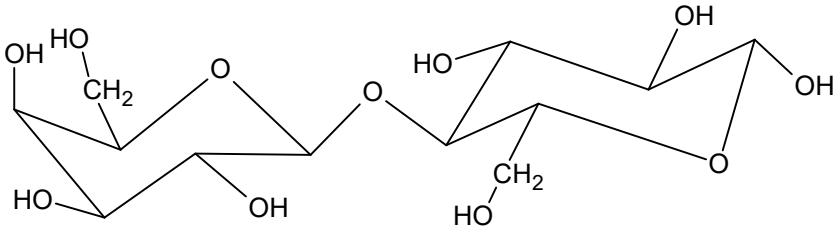
(D) $n = 1 \rightarrow n = 2$

42. Tollen's reagent, basic diaamminesilver(I) solution (in the form of a silver mirror) in the presence of aldehyde, which sugar gives a positive Tollen's test?

I. Sucrose



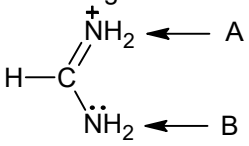
II. Lactose



(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

43. What is the concentration of chloride ion in a solution formed by mixing 150 ml of a 1.50 M NaCl solution with 250 ml of a 0.75 M $MgCl_2$ solution?
(A) 0.563 M
(B) 1.03 M
(C) 1.50 M
(D) 2.25 M

44. Shown below is a Lewis structure of the formamidinium ion. What geometry is exhibited by the two nitrogen in a formamidinium ion



	Geometry of A	Geometry of B
(A)	Trigonal planar	Trigonal planar
(B)	Trigonal planar	Trigonal pyramidal
(C)	Tetrahedral	Trigonal planar
(D)	Trigonal pyramidal	Tetrahedral

45. Molecules from which class of biopolymers can react with water, in the presence of suitable enzyme, to form smaller examples of that class of biopolymers?
I. Protein
II. Polysaccharide
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

46. When solution of barium hydroxide is mixed with a solution of iron(III) chloride, what is observed?
(A) Precipitation of coloured solid
(B) Precipitation of colourless solid
(C) Evolution of colourless gas
(D) Neither precipitation nor gas evolution

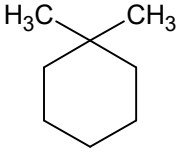
47. NF_3 has bond angle 102.5°, while PF_3 has a bond angle of 96.3°. What is the best explanation for the larger bond angle in NF_3?
(A) The nitrogen 2s-orbital participates more in bonding then does the phosphorous 3s-orbital.
(B) Nitrogen is more electronegative than phosphorous
(C) NF_3 has no unpaired electron while PF_3 has two unpaired electron.
(D) NF_3 is an ionic compound while PF_3 form covalent bond.

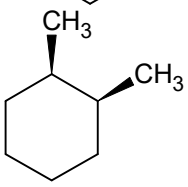
48. What is the role of the acid in the Fischer esterification reaction given below?

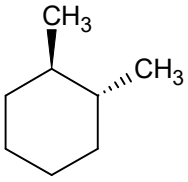
$$CH_3-CH_2-OH+CH_3-CH_2-COOH \rightleftharpoons CH_3CH_2COOCH_2-CH_3+H_2O$$

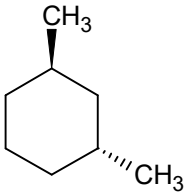
(A) Shift the equilibrium in the right hand direction
(B) Neutralize the base formed as a side product in the reaction
(C) Convert ethanol to a more reactive nucleophile
(D) Convert propanoic acid to a more reactive electrophile.

49. Which molecule has non-zero dipole moment
(A) BF_3
(B) CF_4
(C) NF_3
(D) SF_6

50. In which dimethyl cyclohexane both the methyl groups are on equatorial position in its most stable chair confirmation
(A) 

(B) 

(C) 

(D) 

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. Equal volumes of 0.50 M HCOOH and 0.10 M HCOONa are mixed. What is the pH of the resulting solution? (Given $pK_{a\,HCOOH} = 3.75$ and log5 = 0.7)

52. 0.2495 g of $CuSO_4.5H_2O$ is dissolved in enough water and volume made up to 100 ml. 100 ml of this solution was treated with excess of KI and liberated I_2 required 25 ml of $Na_2S_2O_3$ solution. Find the concentration of $Na_2S_2O_3$.

53. At 800 K, ΔG° for the reaction
$NiO(s) + H_2(g) \rightleftharpoons Ni(s) + H_2O(g)$
is **–** 8992 cal. Find the ratio of pressures of water vapour and hydrogen in equilibrium with NiO and Ni at 800 K in terms of logarithm. [R = 2 cal mol^{-1} K^{-1}]

54. A solution containing 10.2 gm of glycerene per liter is found to be isotonic with a 2% solution of glucose. The molecular weight of glycerene is (Mol. Wt. of glucose = 180 gm)

55. Vapour density of N_2O_4 which dissociate according to the equation
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 25 at $100^\circ C$ and a pressure of 1 atm. The degree of dissociation of N_2O_4 is……?

56. A 1.0 ml solution containing 0.240 μCi of tritium is injected into a Dog's blood stream. After a period of time to allow the isotope to be dispersed, a 1.00 ml sample of blood is drawn. The radioactivity of this sample is found to be 3.2×10^{-4} μCi/ml. The total volume of blood in the Dog in litre is…..?

57. How many stereoisomers are possible for the complex ion $\left[Co(NH_3)_2(CN)_2Cl_2\right]^-$

58. The reaction,
$A \longrightarrow C + D$
was found to be zero order w.r.t. A. The rate constant for the reaction was determined to be $k = 2.5$ mole / ℓ / S. If the initial concentration is 0.5 mole then the value of $t_{1/2}$ is

59. For the transition metal with the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$, how many unpaired electrons are present in its +2 ion in the ground state?

60. 0.2 g of a chloride of an element was dissolve in water and then treated with excess of silver nitrate solution resulting in the formation of 0.50 g of silver chloride. The equivalent weight of the element is

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Sum of values of k for which the lines $3x - y + 2 = 0$, $x - y + 1 = 0$ and $kx - 5y + 6 = 0$ do not form a triangle is
(A) 15
(B) 20
(C) 27
(D) $\frac{47}{2}$

62. The sum of the series $\frac{3}{1^3} + \frac{5}{1^3+2^3} + \frac{7}{1^3+2^3+3^3} + ...$ upto 10 terms is
(A) $\frac{396}{121}$
(B) $\frac{99}{25}$
(C) $\frac{143}{36}$
(D) $\frac{480}{121}$

63. The value of $\sum_{r=0}^{23} {}^{23}C_r \sin^2\left(\frac{r\pi}{2}\right)$ is equal to
(A) 2^{23}
(B) 2^{22}
(C) $2^{23} - 1$
(D) $2^{22} - 1$

64. If α is the angle which each sides of a regular polygon of 10 sides subtends at its centre, then $\cos\alpha + \cos 2\alpha + \cos 3\alpha + ... \cos 9\alpha$ is equal to
(A) 0
(B) 1
(C) –1
(D) $-\frac{1}{2}$

65. Tangents are drawn from point P(–1, k) to the circle $x^2 + y^2 = 4$. The chord of contact of these tangents touch the parabola $y^2 = 36x$, then value of k can be
(A) $\frac{3}{2}$
(B) $\frac{2}{3}$
(C) $\frac{4}{9}$
(D) 36

66. If three normals can be drawn to the parabola $y^2 = 8x$ from the point (k, 0) and if the two normals other than the axis of the parabola are perpendicular to each other then k is equal
(A) 2
(B) –2
(C) 6
(D) 12

67. Number of solution of the equation $\tan 2x\left(1+\sqrt{3}\tan 4x\right) = \sqrt{3} - \tan 4x$ in the interval $[0, \pi]$ is
(A) 6
(B) 5
(C) 4
(D) 1

68. Let $\vec{a} = 2\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} - \hat{k}$. If c is a non-zero vector such that $\vec{a}\cdot\vec{c} = 3|\vec{c}|$ and $|\vec{b}\times\vec{c}| = \sqrt{212}$, then $\vec{c}\cdot\vec{b}$ can be equal to
(A) –2
(B) –6
(C) 4
(D) –14

69. The image of the line segment joining the point A(2, 1, 3) and B(2, 4, 4) in the plane $2x + y - 3z - 10 = 0$ lies on the line
(A) $\frac{x-6}{0} = \frac{y-3}{3} = \frac{z+3}{-3}$
(B) $\frac{x-6}{0} = \frac{y-3}{3} = \frac{z+3}{1}$
(C) $\frac{x-6}{1} = \frac{y-3}{4} = \frac{z-1}{5}$
(D) $\frac{x-6}{1} = \frac{y-6}{3} = \frac{z+2}{1}$

70. If θ_1 and θ_2 be respectively smallest and the largest values of is θ in $(0, 2\pi) - \{\pi\}$ which satisfy the equation $2\cot^2\theta - 5\,\text{cosec}\,\theta + 4 = 0$, then $\tan(\theta_2 - \theta_1)$ is equal to
(A) $\frac{1}{\sqrt{3}}$
(B) $-\frac{1}{\sqrt{3}}$
(C) $-\sqrt{3}$
(D) $\sqrt{2}-1$

71. Coefficient of x^{14} in $\left(1+x^2\right)^{12} + x^2\left(1+x^2\right)^{11} + x^4\left(1+x^2\right)^{10} + ... + x^{24}$ is
(A) ${}^{13}C_7$
(B) ${}^{13}C_6 - 1$
(C) ${}^{12}C_8 - 1$
(D) ${}^{13}C_8$

72. Let y = y(x) be the solution curve of the differential equation $dy(y^2 - 2xe^y) = e^y dx$ satisfying y(0) = 1. This curve intersects the x-axis at a point whose abscissa is
(A) 2 + e
(B) –e
(C) 2
(D) 2 – e

73. In a company, there are seven machines and the probability of any of them to be out of service on a day is $\frac{1}{3}$. If the probability that atmost 3 machines will be out of service on the same day is $\frac{16k}{3^7}$, then k is equal to
(A) 103
(B) 113
(C) 105
(D) 78

74. The mean and standard deviation of 12 observation are 30 and 6 respectively. Each of these 12 observations is multiplied by k and then reduced by m (where $k \neq 0$ and $m \neq 0$). If the new mean and new standard deviation becomes one third of its original values, then m is equal to
(A) 20
(B) –10
(C) –20
(D) –100

75. Let the line y = mx and the ellipse $9x^2 + 4y^2 = 36$ intersect at a point P in the first quadrant. If the normal at P meets the coordinate axes at (k, 0) and $\left(0, \frac{5}{9}\right)$, then k is equal to
(A) $\frac{4\sqrt{2}}{3}$
(B) $\frac{13}{4\sqrt{2}}$
(C) $\frac{10}{9}$
(D) $\frac{5\sqrt{2}}{3}$

76. $\int \frac{15x^{15} + 7x^{13}}{(3x^2 + x^7 + 1)^3} dx$ is equal to
(A) $\frac{1}{2(1+3x^2+x^7)} + c$
(B) $\frac{x^7}{2(1+3x^2+x^7)^2} + c$
(C) $\frac{x^{14}}{2(1+3x^2+x^7)^2} + c$
(D) $\frac{x^{28}}{4(1+3x^2+x^7)^4} + c$

77. The director circle of a hyperbola is $x^2 + y^2 - 4y = 0$. If one end of the transverse axis of the hyperbola is (3, 2), then eccentricity of the hyperbola is

(A) $\frac{\sqrt{14}}{\sqrt{5}}$

(B) $\sqrt{2}$

(C) $\frac{4}{3}$

(D) $\frac{\sqrt{14}}{3}$

78. The area of the region, enclosed by the circle $x^2 + y^2 = 20$ which is not common to the region $y^2 = 8x$ and the line $y = 2x$ is

(A) $20\pi + \frac{4}{3}$

(B) $20\pi - \frac{8}{3}$

(C) $2\sqrt{5}\pi - \frac{4}{3}$

(D) $20\pi - \frac{4}{3}$

79. $\lim\limits_{x\to 1} \frac{\int_{1/x}^{\sqrt{x}} \cos(t^4)dt}{x^2 - 4x + 3}$ is equal to

(A) $\frac{-3\cos 1}{4}$

(B) $\frac{\cos 1}{4}$

(C) $\frac{-\cos 1}{4}$

(D) $\frac{-1}{4}$

80. If α is a repeated root of the equation $x^3 + 4x^2 - 3x + k = 0$, then integral value of k is

(A) 18

(B) –18

(C) 72

(D) –72

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. Let $L = \frac{n^2}{(n+1)(n+2)(n+3)} + \frac{n^2}{(n+2)(n+4)(n+6)} + \frac{n^2}{(n+3)(n+6)(n+9)} + ... \frac{1}{24n}$, then $e^{\lim\limits_{n\to\infty} 2L}$ is equal to

82. $\sum_{k=1}^{12}(1+3+5+.....\ 2k-1)$

83. If c is a point at which Rolle's theorem holds for the function $f(x)=\log_e\left(\frac{x^2+k}{3x}\right)$ in the interval [1, 3], where $k \in R$, then c^2 is equal to

84. $\left(\cos^{-1}x\right)^2+\left(\cos^{-1}y\right)^2+2\left(\cos^{-1}x\right)\left(\cos^{-1}y\right)=4\pi^2$, then $x^2 + y^2$ is equal to

85. The sum of the slopes of all the common tangents of the circles $x^2 + y^2 – 4x + 2y – 4 = 0$ and $x^2 + y^2 – 12x –4y + 36 = 0$ is

86. If the system of linear equations $x + 2y + kz = m$, $x + y + z = 6$ and $3x + 2y + z = 14$ has more than 2 solutions, then $m – k$ is equal to

87. Let f(x) be a polynomial of degree 5 such that x = 1, x = 2 are its critical points. If $\lim_{x\to0}\left(3+\frac{f(x)}{x^3}\right)=7$, then local maximum value of f(x) is

88. Let $A=\left[a_{ij}\right]$ and $B=\left[b_{ij}\right]$ be two 3×3 real matrices such that $b_{ij}=2^{(i+j-1)}a_{ij}$ where i, j = 1, 2, 3. If the determinant of B is 256, then determinant of A is

89. The number of 4 letter words (with or without meaning) that can be formed from the nine letter of the word 'SEGREGATE' is

90. The area (in sq. units) of the region $\{(x, y) \in R^2 : x^2 + 2x \le y – 4 \le –2 – x\}$ is